Dev-log for Final Major Project - Interstellar Adventure

https://cheng-min.itch.io/interstellar-adventure

Cheng Min

Content

Personal Aims

- Craft a captivating **role-playing puzzle-solving** game that seamlessly incorporates **Sokoban** gameplay mechanics, offering players a unique and intellectually stimulating experience.

- Implement a **combined positive and negative feedback loop** intricately woven into the game dynamics, dynamically adapting the difficulty level based on the player's performance and progress.

Game Overview

1. Group Members

Cheng Min	Level Design / Game Design / Narrative Design
Ziwei Su	Game Development / Game Design
Dixuan Xu	2D Artist / Game Design

2. Overview

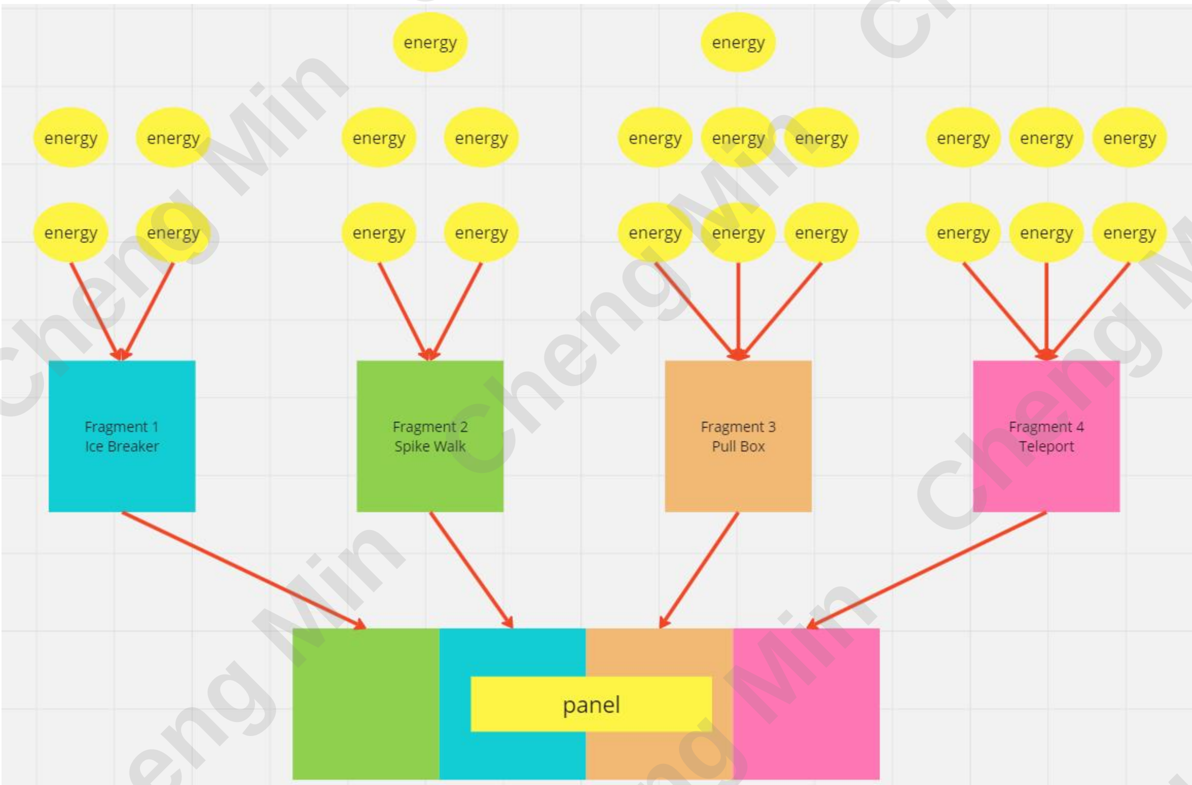
Interstellar Adventure is an RPG Sokoban game that players assume the role of Jack, an extraterrestrial being whose spacecraft has crash-landed on an enigmatic planet. Their mission is to navigate challenges by collecting scattered crystals to repair the damaged spaceship panels, ultimately enabling Jack to embark on a journey back home.

3. Gameplay

As a Sokoban game, Interstellar Adventure integrates RPG elements, enhancing the traditional box-pushing mechanics with role-playing and a narrative. It aims to provide players with a more immersive experience. Furthermore, the incorporation of positive and negative feedback loops into its core game mechanics serves as a pivotal element in managing game difficulty, particularly crucial in skill utilization within the game.

The game's narrative unfolds as the player character, Jack, an interstellar explorer, experiences a spaceship collision with meteorites during a cosmic journey. Forced to make an emergency landing on an unfamiliar planet, Jack discovers that the spaceship's panels have shattered into four fragments. Observing that yellow crystals on the planet can energize and repair the spaceship panels, Jack embarks on a journey to collect these crystals and restore the damaged panels.

In the game, players must solve puzzles through box-pushing mechanics, collecting crystals to progress to the next level. The gathered crystals can be used to charge panel fragments and exchange for skills. The four panel fragments provide players with four distinct skills: Ice Breaker, Spike Walk, Pull Box, and Teleport.



4. Significance

Sokoban games have several distinctive features, including:

- Core Box-Pushing Mechanism: The primary gameplay involves moving boxes to designated locations. Players control a protagonist, often a warehouse worker, to push boxes and complete tasks with specific layouts.
- Simple yet Challenging: Despite simple rules, Sokoban games offer significant challenges. Due to limited space and the relative positions of boxes, players must carefully plan each move.
- Level Design: Sokoban games typically feature progressively challenging levels. Players face increasingly complex layouts and obstacles, requiring continuous improvement in puzzle-solving skills.
- Uniquely Solvable Puzzles: A well-designed Sokoban puzzle should be solvable, meaning there is a solution to move the boxes to their target locations. This ensures each level has a clear resolution.
- Immersive Puzzle Experience: Despite its simplicity, Sokoban games stimulate players' logical thinking and planning abilities, providing a profound and immersive puzzle-solving experience.
- Classic and Popular: Sokoban is a classic puzzle game that has been popular since the 1960s. Its simple yet challenging design has garnered a dedicated player base across various platforms.

While Sokoban games have numerous strengths, there are also some potential drawbacks:

- Limited Gameplay Elements: The core gameplay of Sokoban, centered around pushing boxes, is relatively simple. Players seeking more complexity and diversity in game mechanics may find it lacking in variety.
- High Difficulty Leading to Frustration: As the difficulty of levels increases, some players may find the game overly challenging, leading to frustration. This can make the game less appealing to certain players.

- Lack of Narrative: Sokoban games typically lack a deep narrative, emphasizing puzzle-solving over storytelling. For players who enjoy games with rich narrative elements, this might be considered a drawback.
- Perceived Monotony: Despite each level presenting a new puzzle, the fundamental gameplay may be seen as monotonous over extended periods due to its relative simplicity.

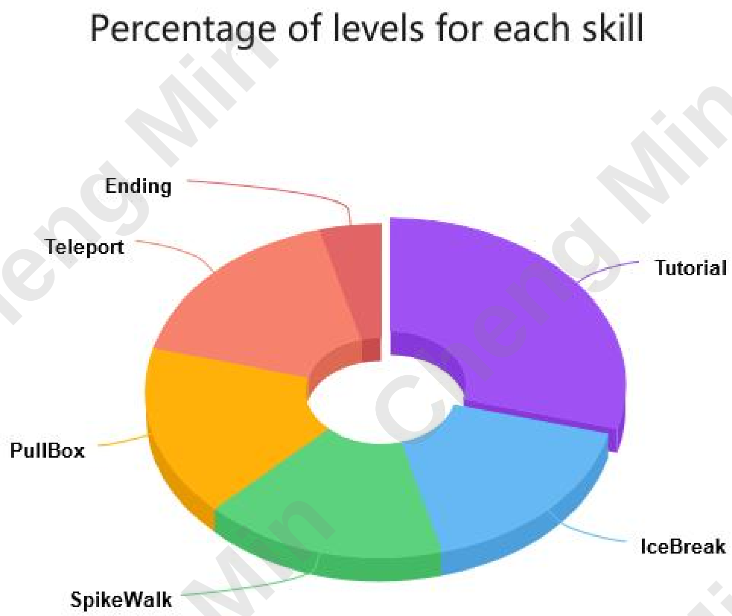
In addressing these issues, we integrated RPG elements into the Sokoban game to enhance narrative depth and introduce skill design. Employing a positive feedback loop amplifies player achievement and minimizes frustration. To prevent skill overuse from disrupting game balance and making it too straightforward, a negative feedback loop was introduced to moderately increase difficulty, ensuring the game maintains an appropriate level of challenge. This approach diminishes the snowball effect of the positive feedback loop, encouraging players to consistently engage in the game's challenges.

Design Rules

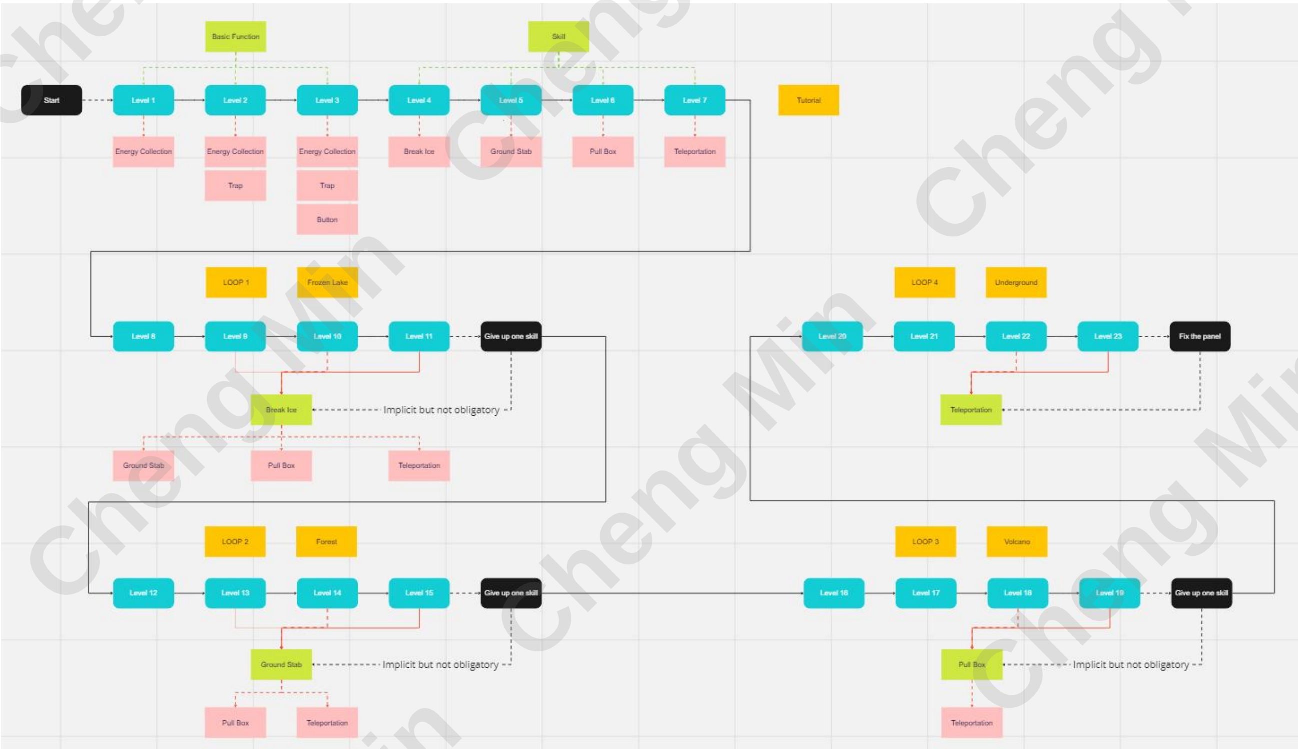
- **Negative feedback stabilizes the game.**
- **Positive feedback destabilizes the game.**
- **Negative feedback can prolong the game.**
- **Positive feedback can end it.**
- **Negative feedback loop sustain engagement and foster competitiveness.**
- **Positive feedback loop provide a sense of progression.**
- **Feedback systems can take control away from the players.**
- **Negative feedback should be indirectness, When it becomes unfair.**
- **Negative feedback provides opportunities for falling behind to victory.**
- **Negative feedback contributes to increased uncertainty in the game's outcomes.**
- **Both feedback loops must be delicately balanced.**

Positive & Negative Feedback loops

1. Game Flow



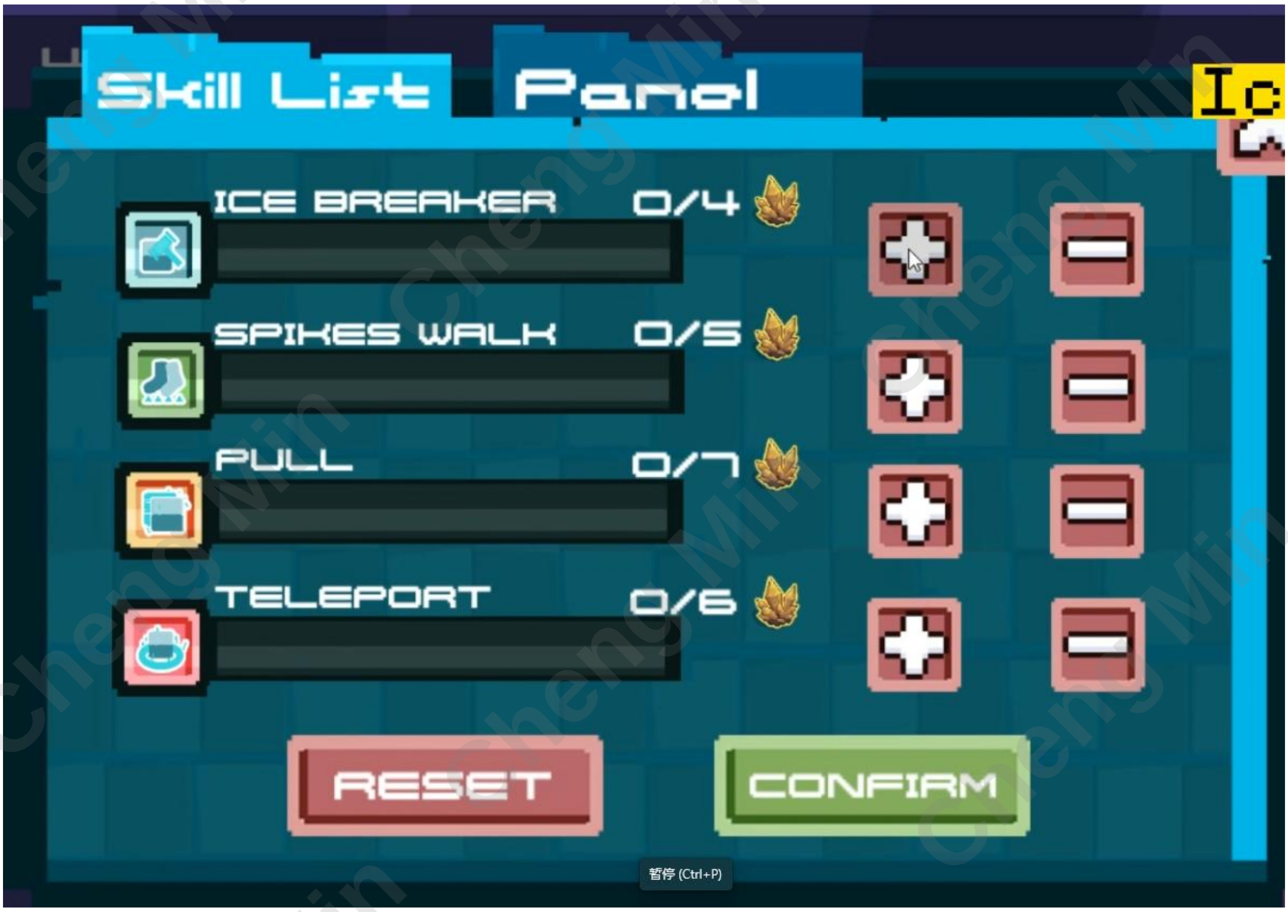
Interstellar Adventure consists of a total of 24 levels, with levels 1-7 serving as tutorial stages. Levels 8-11 are designed for utilizing the Ice Breaker skill, levels 12-15 for the Spike Walk skill, levels 16-19 for the Pull Box skill, levels 21-23 for the Teleport skill, and level 24 serves as the final completion stage for players to conquer.



The final two levels of each major group have been expanded to incorporate more opportunities for skill utilization. The impact of skills on the difficulty of these levels is notably significant.

2. Positive Feedback loops

To prevent player frustration due to excessively challenging puzzles, I've designed special skills available for players. These skills, obtained by exchanging crystals collected in the game, significantly reduce level difficulty, enhancing player achievement.



Gathering more energy increases the quantity of chargeable fragments.

↓

The increased number of chargeable fragments provides access to a greater variety of skills.

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With the ability to use more skills, the likelihood of successfully completing levels is enhanced.

But these may lead to the following issues:

Snowball Effect	The positive feedback loop might cause a snowball effect, where the more skills a player uses, the easier the game becomes, potentially diminishing overall difficulty.
Reduced Challenge	With the strengthening of the positive feedback loop, the game could become overly easy, diminishing the intrinsic challenge of Sokoban and diminishing the player's sense of accomplishment.
Loss of Puzzle Complexity	Excessive positive feedback may result in puzzles losing their original difficulty and complexity, shifting the focus more toward action or skill usage rather than puzzle-solving.
Diminished Strategic Element	Sokoban games often involve strategic thinking. Introducing too much positive feedback may reduce the need for strategic thinking at both the local and global levels.

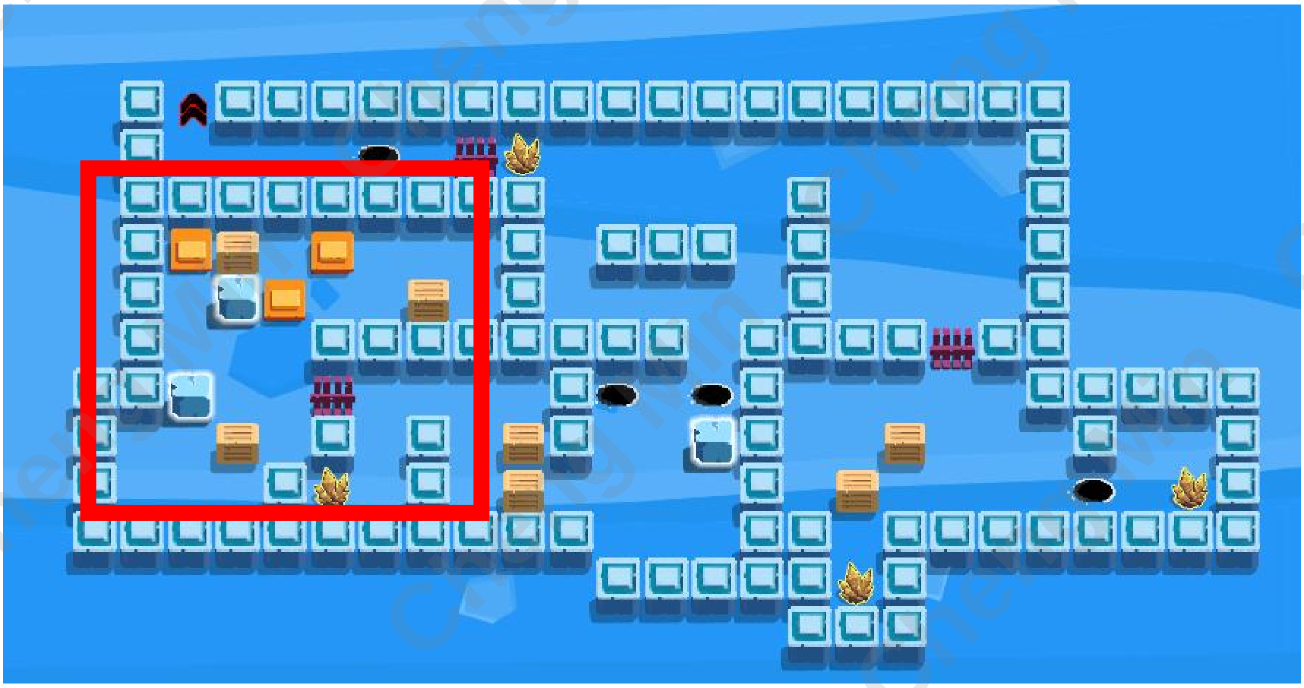
To address these issues, it's crucial to maintain a balance in the overall game experience while introducing skills and positive feedback loops. Gradually increasing the difficulty of skills, incorporating negative feedback loops to counterbalance positive effects, and providing more challenging levels are potential strategies to ensure depth and balance in the gaming experience.

3. Negative Feedback loops

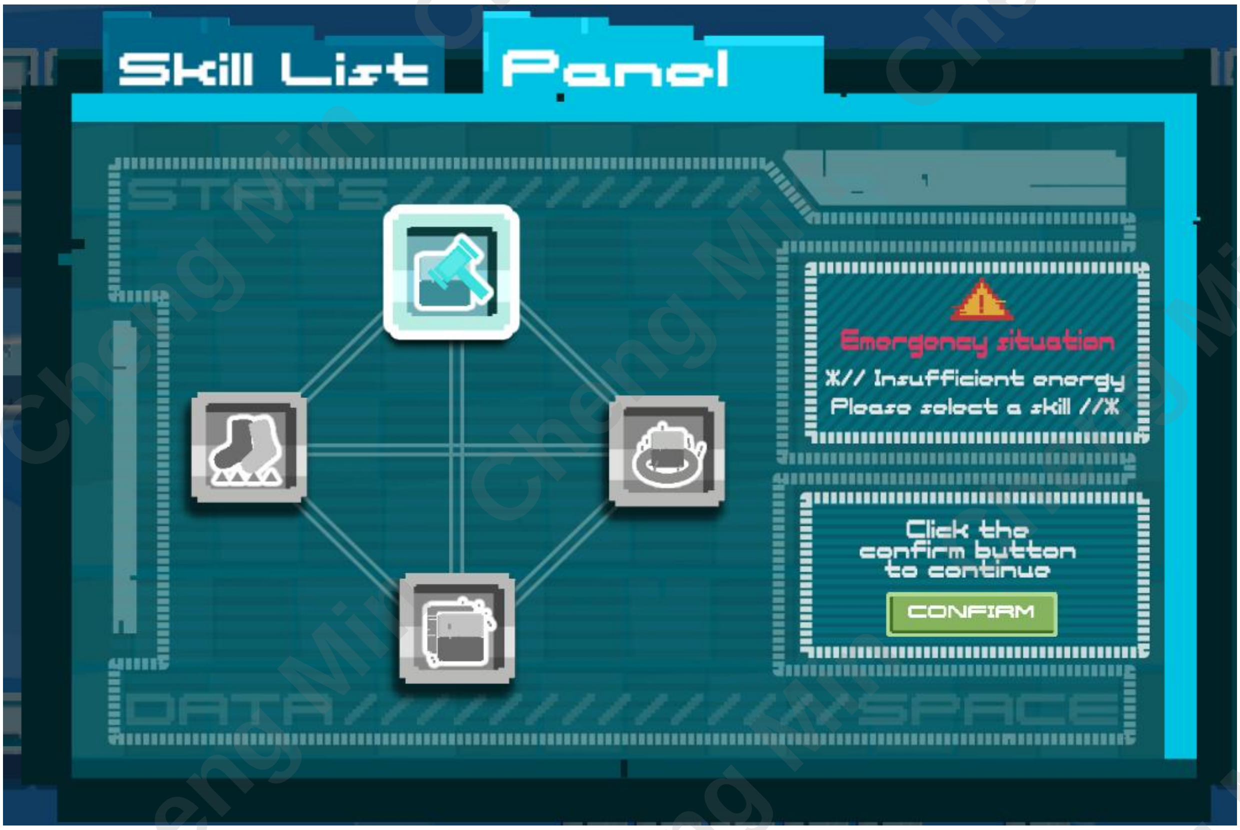
After excessive misuse of skills, the game might become too easy, diminishing the player's interest in the challenge. Therefore, I introduced a negative feedback loop to limit the snowball effect caused by the positive feedback loop. For example, every four levels form a set, corresponding to four skills. The last 1-2 levels in each set pose significant challenges to counterbalance skill usage.

Moreover, as the game unfolds, players progressively acquire proficiency in gaming techniques. Even when employing skills, the overall design of the 24 levels systematically intensifies in complexity, ensuring players consistently find joy in confronting challenges.

For example, in Level 11, the primary design focuses on the use of the Ice Breaker skill. The positions of the two ice blocks, as shown in the following diagram, are crucial in this map. Without breaking these two ice blocks, players would need approximately 95 steps to complete the puzzle with the three boxes on the left. However, with the use of the skill, it only requires around 43 steps for completion.



Most notably, we introduced the option to passively relinquish skills in the game. At the game's outset, we offer players four skills to choose from. However, as the game progresses, players receive prompts like "Spaceship energy depleted, replenish by sacrificing one panel fragment." This means players must permanently surrender skill usage by choosing a fragment with a skill after completing each set.



This design also serves to mitigate the excessive positive feedback from skill usage. As players become more adept and powerful in the game, the available skills gradually diminish. Consequently, the means by which players can solve puzzles also decrease, ensuring a steady increase in the game's difficulty. In the later stages of the game, players will lose all skills and become unable to use them.

The availability of skills in the game decreases gradually as players progress.

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Due to the reduction in skills, the means by which players can solve puzzles also decrease.

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The reduction in available skills leads to a steady increase in the game's difficulty.

This design may sound counterintuitive and could potentially lead to the following issues:

Reducing Player Motivation	Excessive negative feedback may lead to players losing interest and motivation, diminishing their desire to continue challenging the game.
Diminishing Game Attractiveness	If negative feedback is too intense, it may cause players to abandon the game prematurely, missing out on subsequent engaging content and experiences.
Impact on Player Experience	Excessive negative feedback may affect the overall gaming experience, making it unpleasant and consequently lowering players' overall perception of the game.
Hindering the Learning Curve	Intense negative feedback can steepen the game's learning curve, making it challenging for players to adapt and master the game's mechanics, thus affecting their engagement.

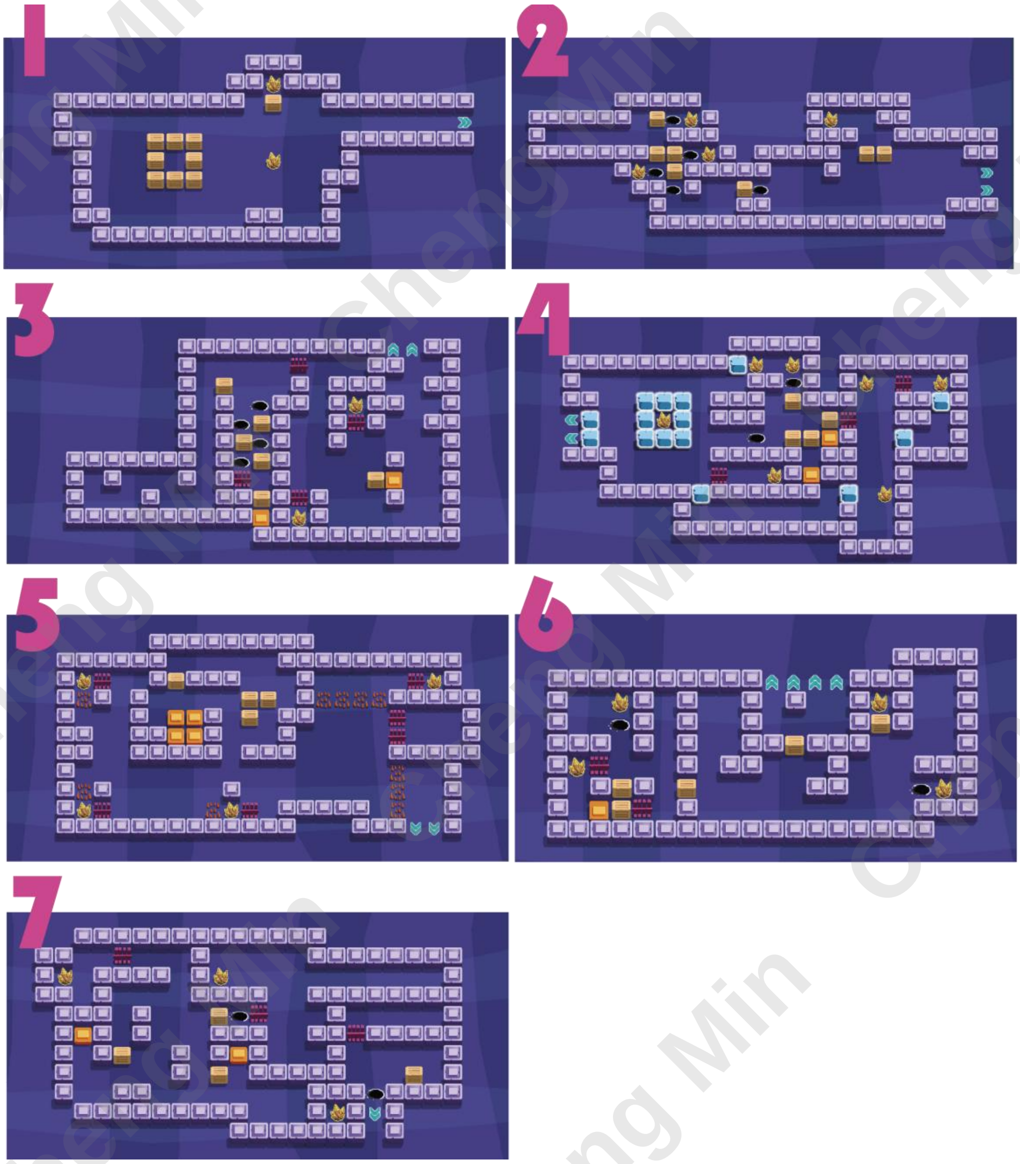
As a result, we crafted a logically consistent backstory: each submitted fragment represents a repaired segment of the spaceship's panel. The player's objective is to restore the four panel fragments and ultimately power up the spaceship. When the negative feedback loop aligns with the game's purpose, it becomes reasonable and more easily accepted.

Level Design

1. Tutorial

The tutorial spans levels 1 to 7 in the game. Beyond instructing directional controls on the initial interface, the first level introduces the undo and refresh keys, emphasizing the importance of crystal collection. The second level introduces the concept of holes and how to manage them: by pushing boxes into them. The third level focuses on button and gate mechanics, familiarizing players with their usage.

In these levels, obstacles have been strategically placed at the exit points. Examples include ice blocks at the exit in level 4, spikes in level 5, a blocked path by boxes in level 6, and a route accessible only through teleportation in level 7.



Text-based dialogue guidance is also provided to assist players in understanding and implementing these skills.

Level 1:
"The spaceship's panels are shattered into four pieces..."
"I must repair them to leave this place."
"Those yellow crystals are glowing; they seem to be useful."
"(Reset button) If needed, use the button in the upper right to refresh the current level.
"Alternatively, you can press the 'Z' key to undo the last move."
"Let's continue down."

Level 2:
"There are some holes on the ground, perhaps meteorite craters?
"I need to figure out how to get past."

Level 3:
"Those orange buttons control the gates."

Level 4:
"Some paths are blocked by ice; I remember the first-panel fragment has the skill to break it."
"(skill list) It's time to open the skill list. "
"Insert 4 crystals into the first ice-breaking skill, and I can use it directly."

Level 5:
"I can't walk through these thorn bushes..."
"Unless, I use the skill 'spike walk' from the second-panel fragment."

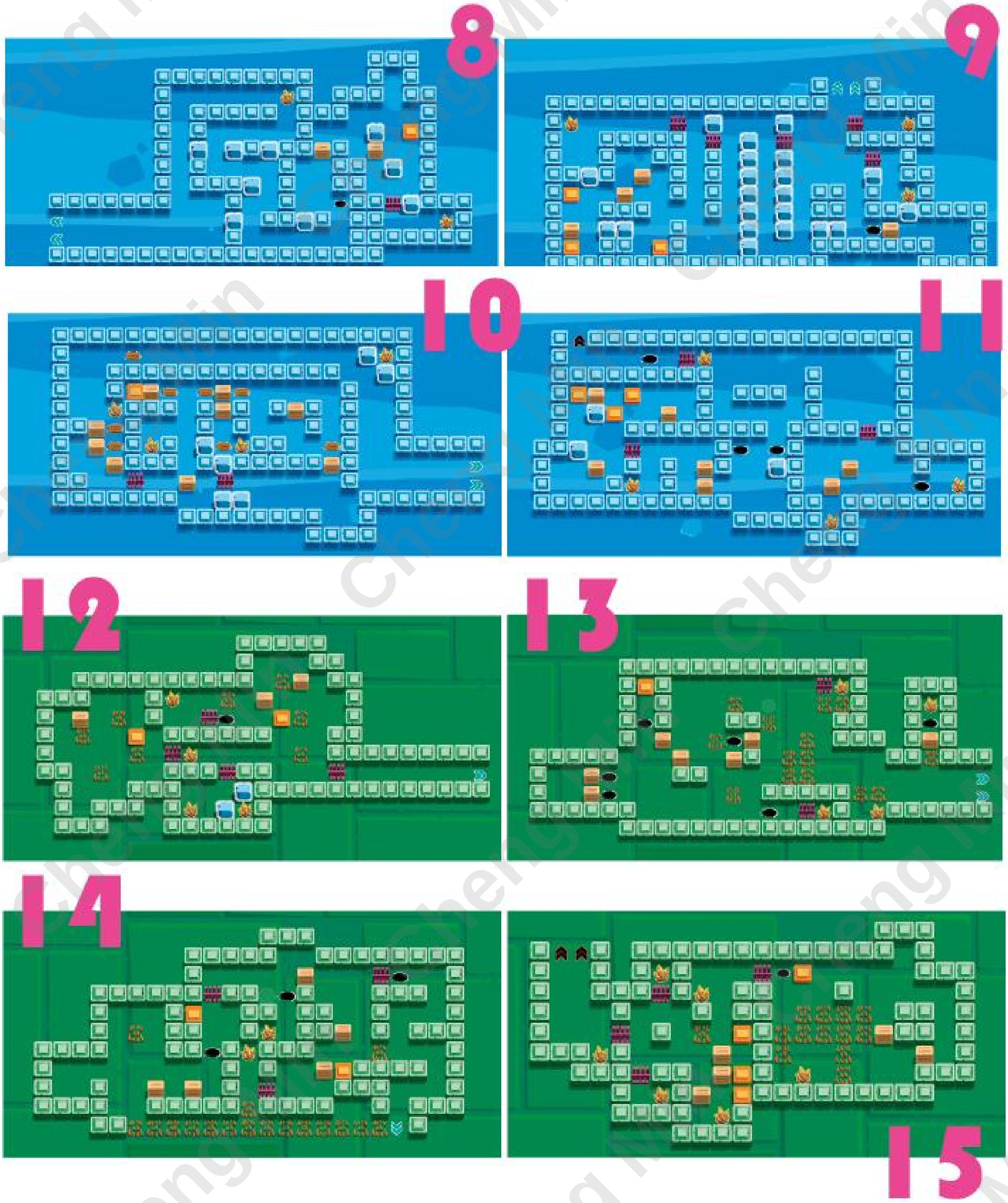
Level 6:
"Some boxes seem impossible to push."
"But what if I can pull them..."
"The third-panel fragment allows me to pull them, but it requires 7 crystal energy."
"Press the 'K' key and arrow keys when using the skill."

Level 7:
"No ice, no spikes, and no boxes to pull; these sturdy walls are impenetrable."
"It's time to use the last skill; I can teleport to two different locations."
"Insert six crystals into the fourth skill 'teleport,' and press the 'L' key and arrow keys to use."

Level 8:
"When facing insurmountable challenges..."
"exchanging collected crystals for skill usage can be helpful."
"However, the spaceship needs some energy to maintain its dormant state; "
"It will prompt in the panel when required."

2. Skill Level

The levels from 8 to 23 are primarily designed for different skills, with each skill having four associated levels. The crucial puzzle-solving positions in these levels are related to the use of specific skills. The decision to use skills or not significantly impacts the outcome of each level.



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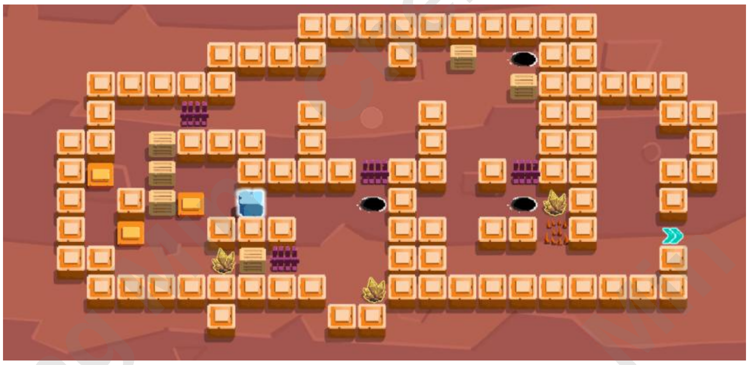
3. Skill submit

By arranging the level sequence, aligning the appearance order of obstacles with corresponding skills, and using different background colors for levels, we subtly suggest to players to submit skills in the order of ice breaker, spike walk, pull box, and teleport. This sequence provides the optimal gaming experience. Unfortunately, designers can't always predict player behavior. To ensure that players who do not follow this sequence can still enjoy the thrill of using skills in subsequent experiences, I introduced additional opportunities for skill use in later levels.



For example, in level 18, besides the pull box skill helping players solve puzzles more efficiently, I strategically placed an ice wall that can be broken at a crucial position. If a player still possesses the ice breaker skill by the time they reach level 18, using it will allow them to smoothly navigate and complete the level.

In level 19, spikes are introduced, allowing players to collect six crystals and proceed to the next level without moving any boxes. However, using the spike walk skill requires spending six crystals, balancing the gained and spent resources. Players need to consider whether this trade-off is worthwhile.

Level 21 provides opportunities for using both the ice breaker and spike walk skills. Regardless of which skill the player has in hand, it can effectively assist in quickly completing the level. Of course, having the teleport skill would be even better.

4. Difficulty Grading

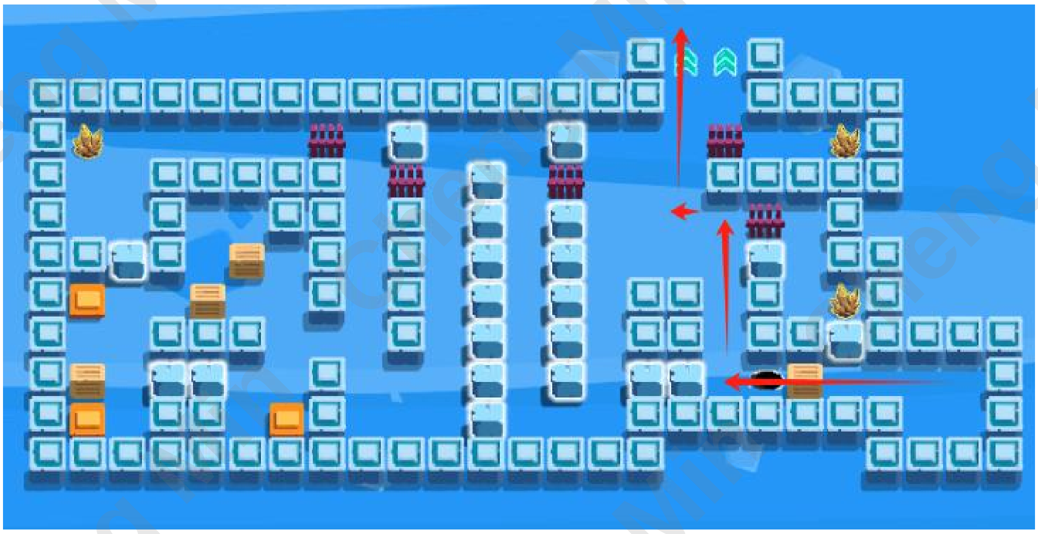
The current version of the game does not include difficulty levels for players to choose from, but within individual levels, various ways to solve them are provided.

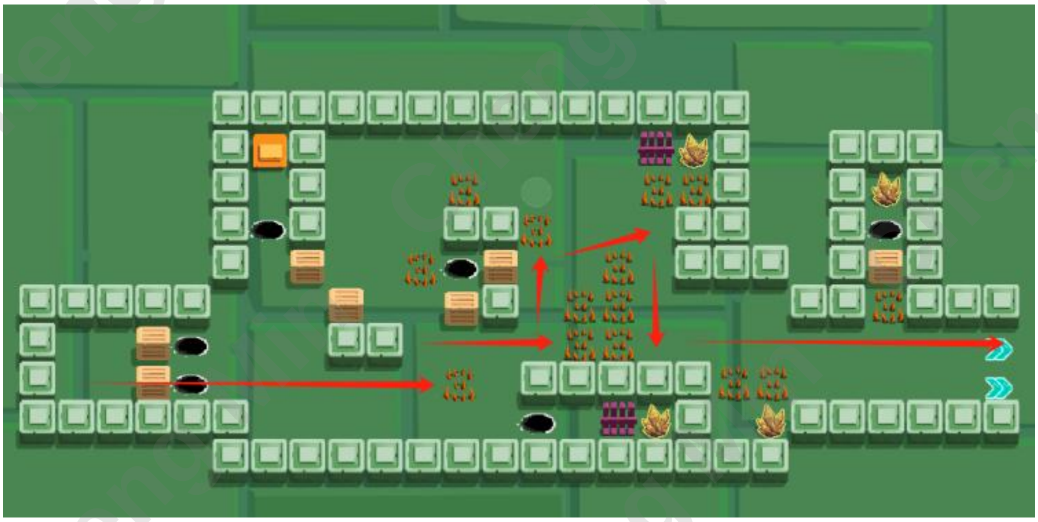
Standard mode

Completing the game according to the game guide involves straightforward puzzle-solving in simple levels by pushing boxes, while in challenging levels, players have the option to selectively use skills for completion

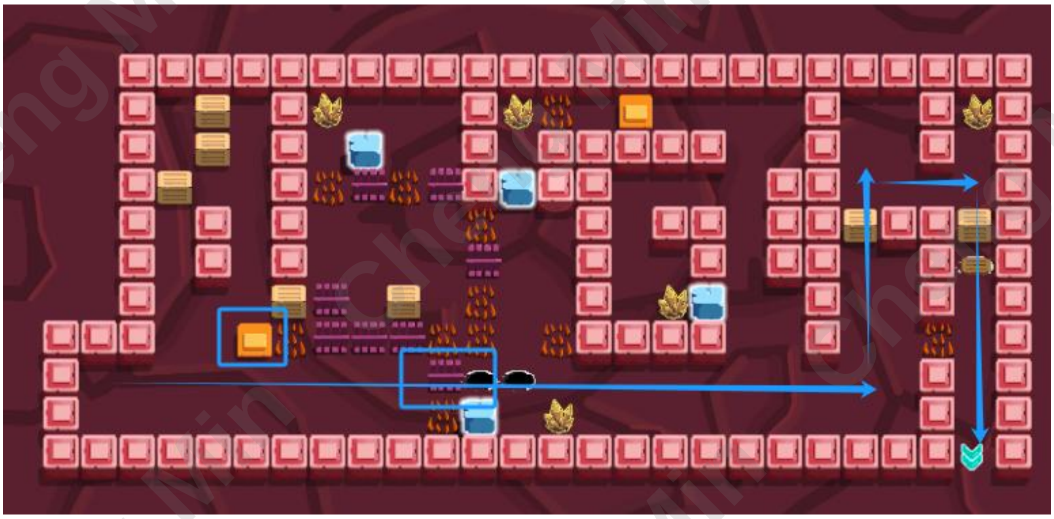
Easy Model

In some challenging levels, players have the option to abandon the collection of crystals and proceed directly to the exit. In fact, excluding the crystals from these levels, the crystals collected in other levels are sufficient for players to repair the panel and complete the game.





Additionally, in some levels, players only need to solve a small portion of the puzzle to clear the path to the exit entirely. Of course, if they wish to acquire more crystals, they still need to complete the remaining puzzles.



Hard Model

For players highly skilled in Sokoban-style games, the level design also provides a challenging option. From level 8 to 24, all levels can be completed without using any skills.

Moreover, the crystal collection limit is set at 45, meaning players can collect a maximum of 45 crystals from the beginning to the end without using crystals unless necessary to meet the completion conditions.

Game Version Updates & Playtesting Feedback

Level Iteration Overview

Version 1.0

Story Background

I, the player character, am an android on a cosmic exploration mission, equipped with a spaceship named "Interstellar Explorer," a trusted companion on my interstellar mysteries exploration. We have been navigating the unknown cosmic wonders throughout the universe.

However, one day, our spaceship encountered a dreadful space storm, causing significant damage to the primary energy panels. I found myself in an unavoidable predicament, forced to land on an unfamiliar planet with the hope of repairing the spaceship and resuming our journey.

On this planet, I discovered:

- **Mysterious Forest**: Inhabited by peculiar creatures and plants (obstacles), this forest requires careful exploration to find energy sources.
- **Frozen Lakes**: Covered by thick ice layers, traversing these frozen lakes is necessary to uncover energy hidden beneath the icy surface.
- **Desert Ruins**: In the desolate region of the planet, I stumbled upon abandoned city

ruins covered in dust. Here, I found energy, but I had to navigate the threat of sandstorms to safely retrieve it.

- **Lava Caves**: Venturing into underground caves filled with lava, I faced dangers of high temperatures and lava pools while seeking energy. Carefully avoiding lava flows and magma blocks, I had to swiftly retreat.
- … (Further exploration unfolds)

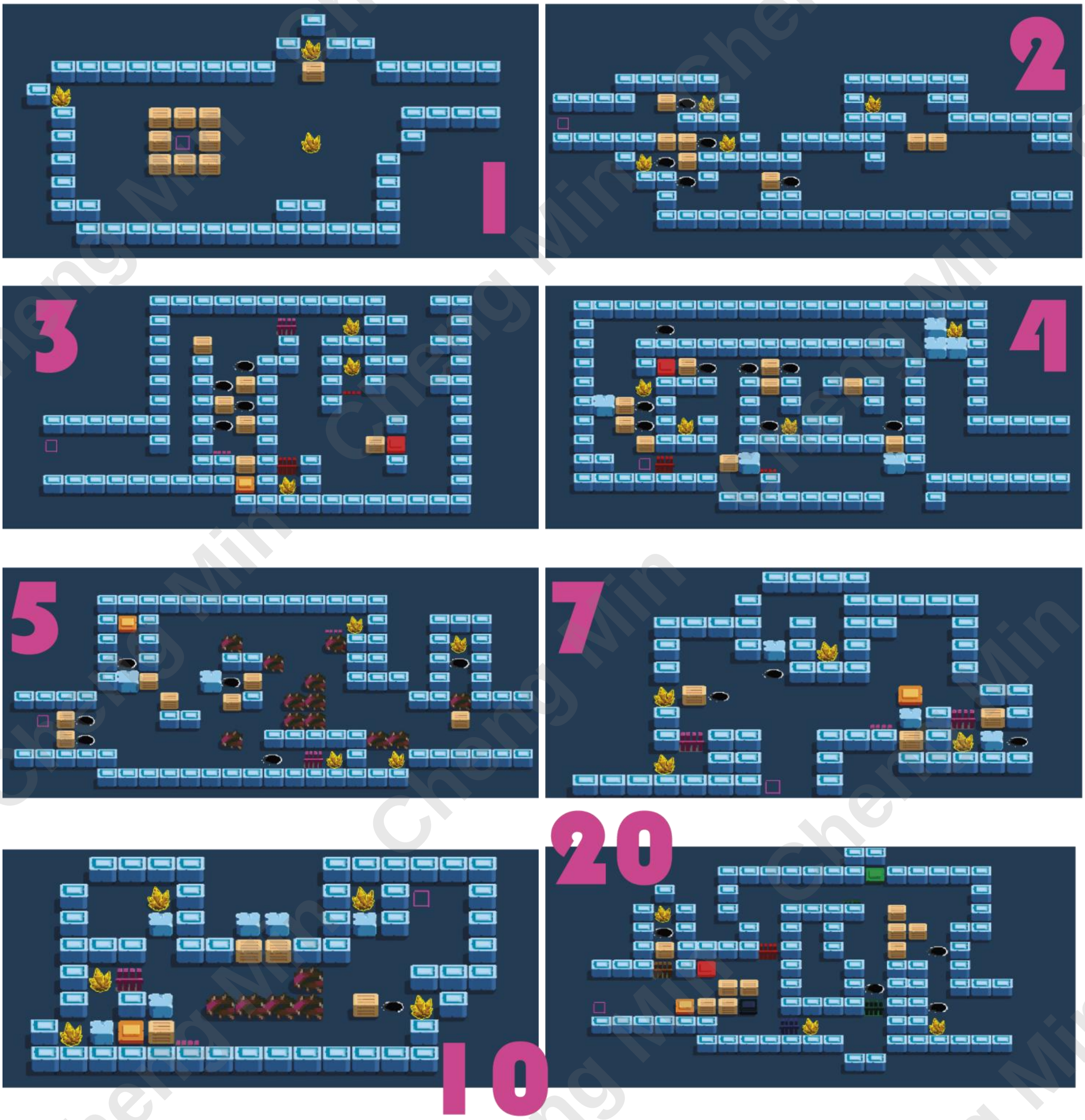
I discovered energy crystals that could provide sufficient power to sustain the spaceship. However, to repair the spaceship, I needed to charge the fragments of the spaceship panel. Analyzing records and sensor data from the fragments, I gradually acquired a special skill to better navigate the unknown planet. However, maintaining the spaceship in standby mode also requires energy. At times, I had to reinstall charged fragments into the panel to ensure the spaceship remained powered.

In the end, I returned to the spaceship, installed all the fragments, and successfully restarted the spaceship. We continue our journey, exploring the unknown cosmos, upholding the mission set by scientists to unravel the mysteries of the universe.

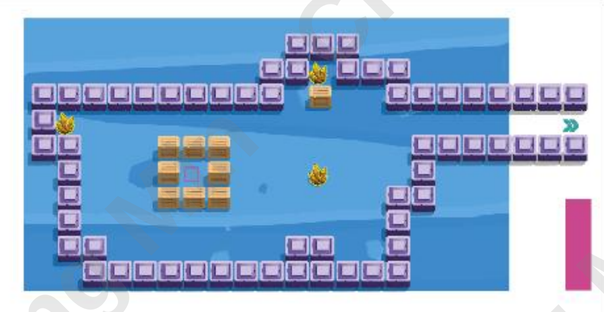
skills

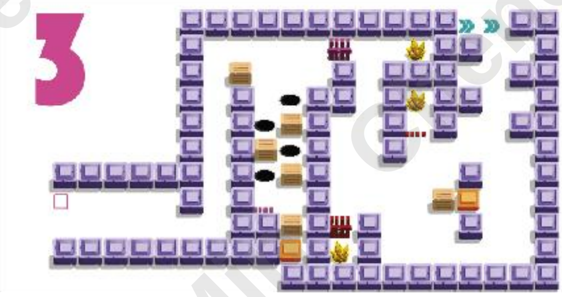
Skill	Background Colour	Notes
Break ice	Blue	This skill enables the player to break the ice obstructing the path.
Ground Stab	Green	This skill allows passage over surfaces covered with stab.
pull box	Orange	With this skill, you can pull boxes in a direction beyond your own.
Teleportation	Red	Using this skill enables instantaneous teleportation to any location on the map.

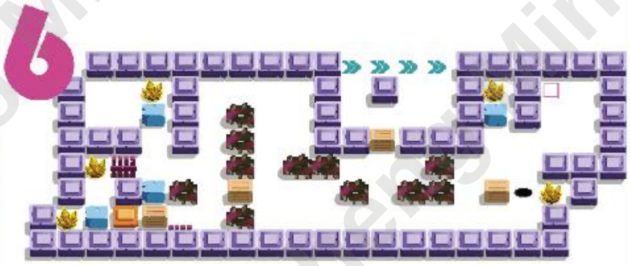
Level Iteration

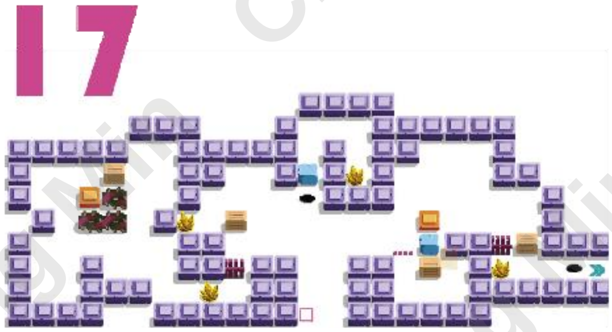


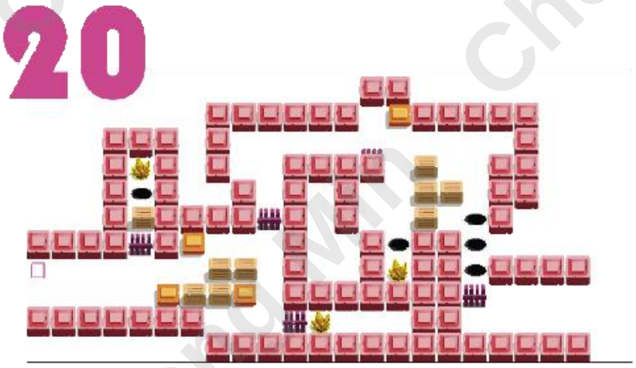
Version 1.1

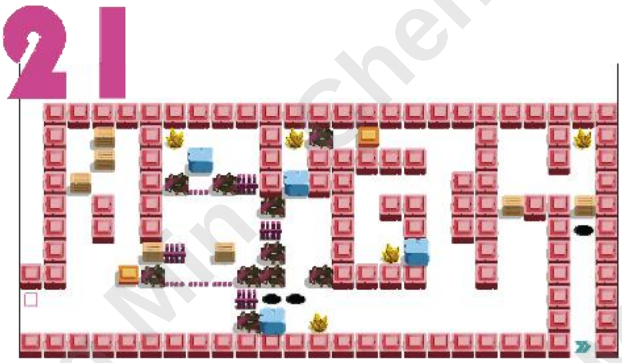










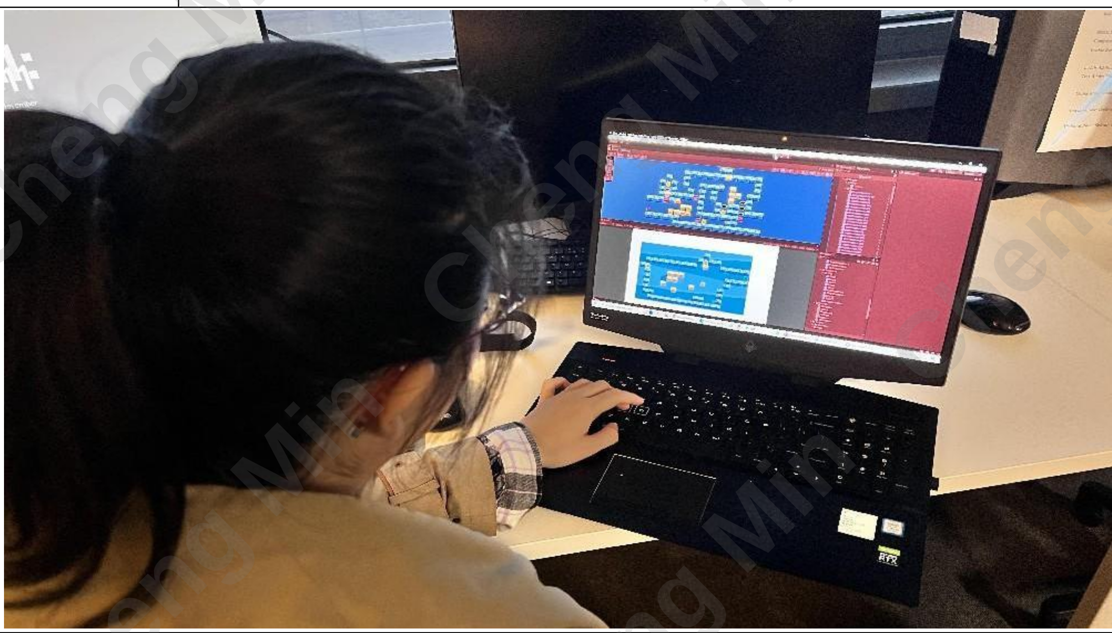


skills

Skill	Background Colour	Notes
Break ice	Blue	This skill enables the player to break the ice obstructing the path.
Ground Stab	Green	This skill allows passage over surfaces covered with stab.
pull box	Orange	With this skill, you can pull boxes in a direction beyond your own.
Teleportation	Red	Utilizing this skill allows instantaneous teleportation to **a position within two grid spaces**.

Playtesting (11.3)

Player 1	**Level 1**	**Level 2**	**Level 3**	**Level 4**
Ruimeng	smoother and easier	Need a little clue to clue the player in on the existence of the hole		The ice and the blue walls look too similar
Comprehensive Feedback	● Consider implementing a save function, allowing players to explore scenarios with different skill usages. ● Consider adding a one-time refresh option for the current level. ● Introduce satisfying moments, such as the exhilarating feeling of breaking an entire stretch of ice walls.			



<table>
<tr><th>Player 2</th><th>Level 1</th><th>Level 2</th><th>Level 3</th><th>Level 4</th></tr>
<tr><td>David</td><td></td><td></td><td></td><td></td></tr>
<tr><td>Comprehensive Feedback</td><td colspan="4">● Consider creating a quick gameplay experience by isolating a few levels, allowing players to have a brief yet engaging experience.
● Alternatively, implement optional levels, enabling players to experience subsequent levels even if they encounter difficulties in certain stages.</td></tr>
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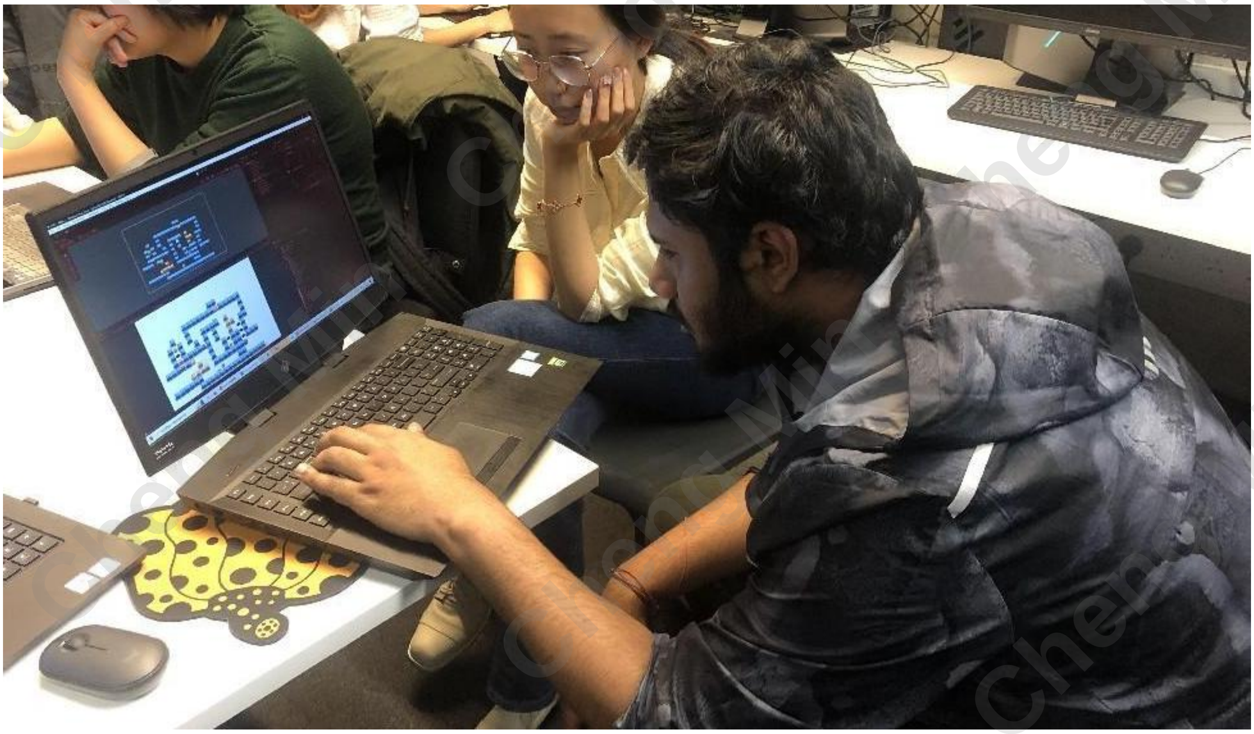
<table>
<tr><th>Player 3</th><th>Level 1</th><th>Level 2</th><th>Level 3</th><th>Level 4</th></tr>
<tr><td>Shan</td><td>Baby level</td><td></td><td>Consider providing a prompt at the beginning to inform players about the number of times they will need to use the ice-breaking skill.</td><td></td></tr>
<tr><td>Comprehensive Feedback</td><td colspan="4">Consider adding a timer and a final ranking system to enhance the overall gaming experience.</td></tr>
<tr><td colspan="5"></td></tr>
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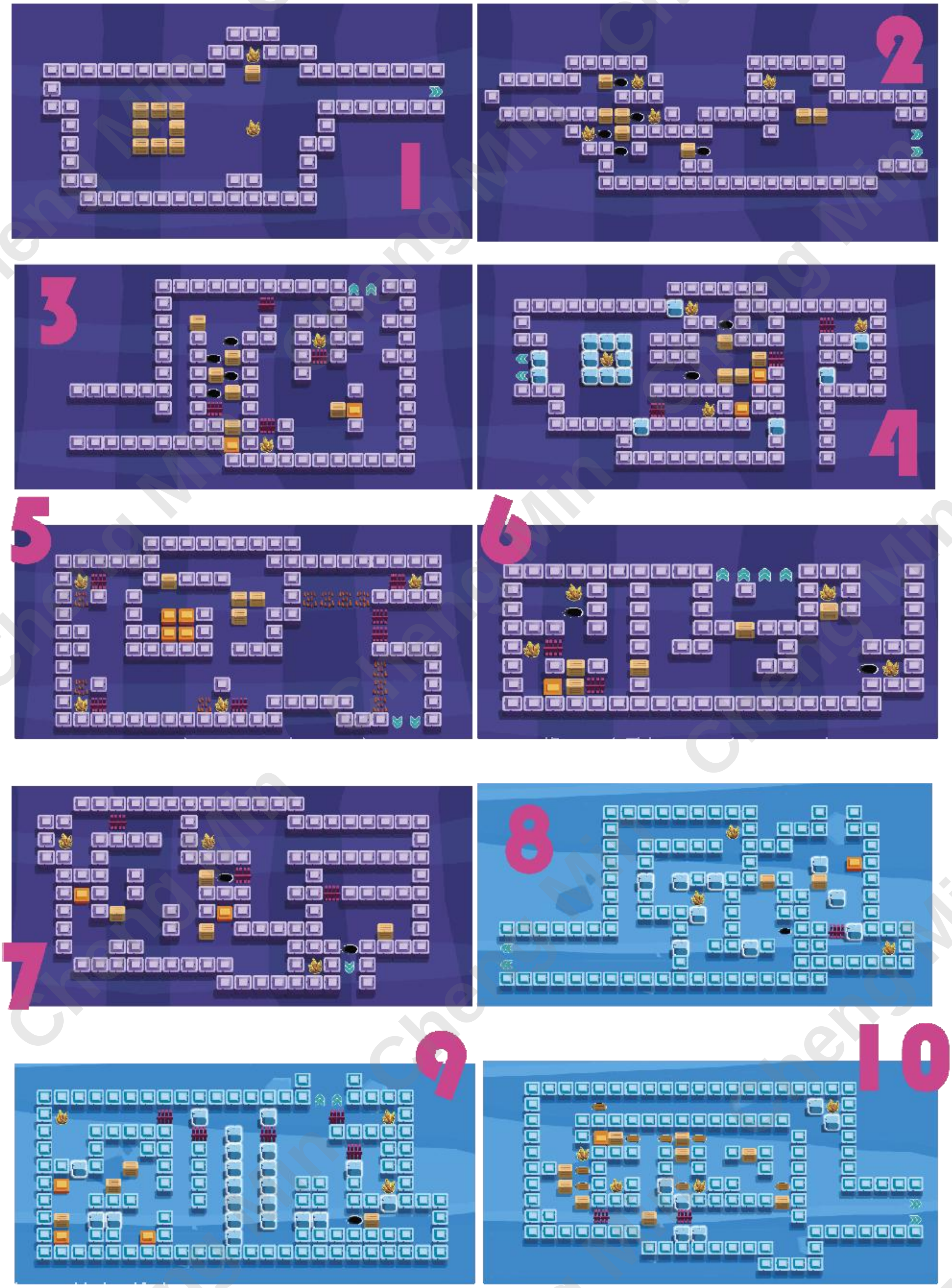
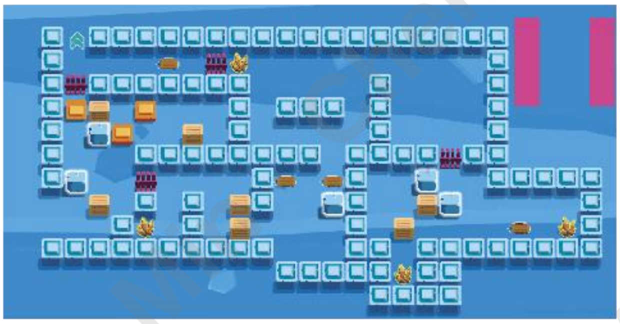
Version 2.0 - 2.1

Level Iteration

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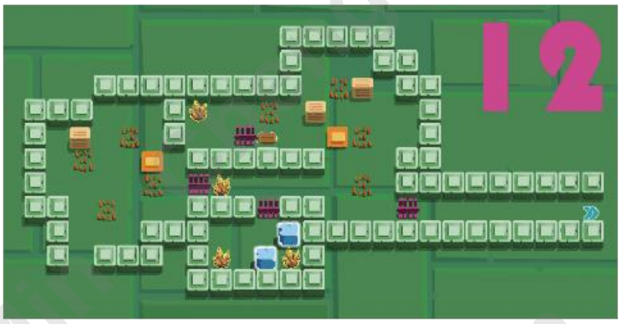
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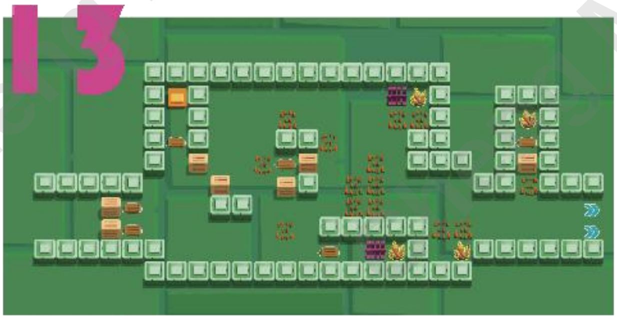
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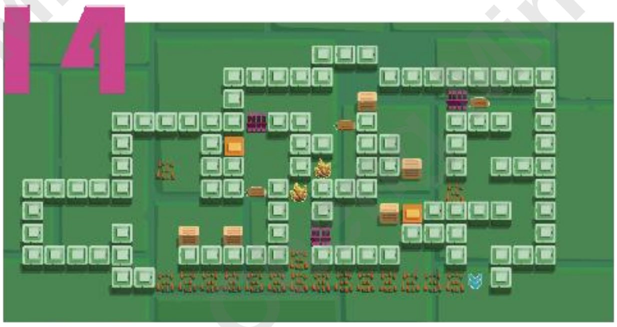
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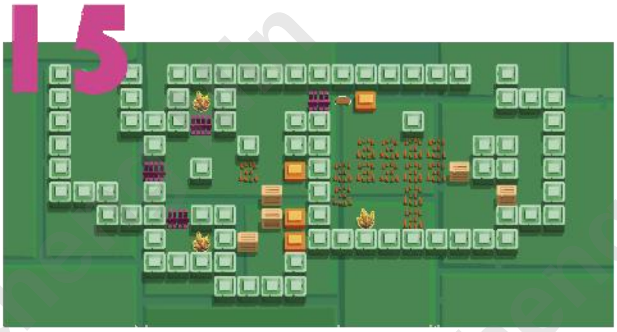
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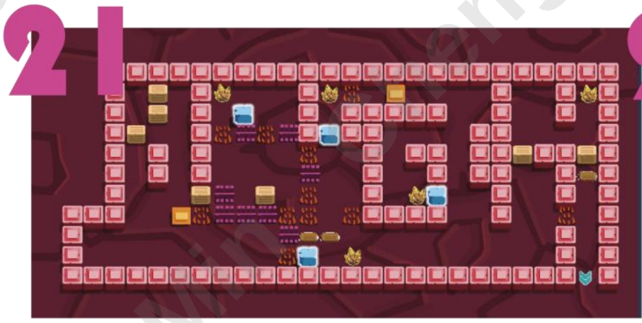
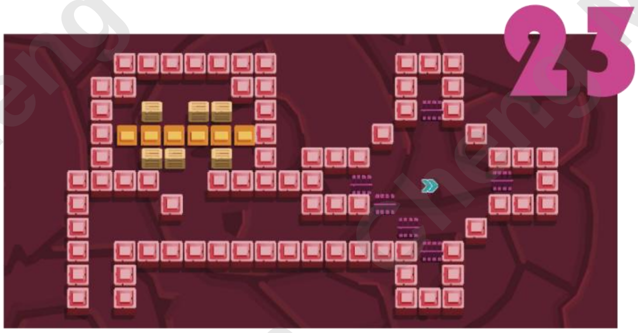


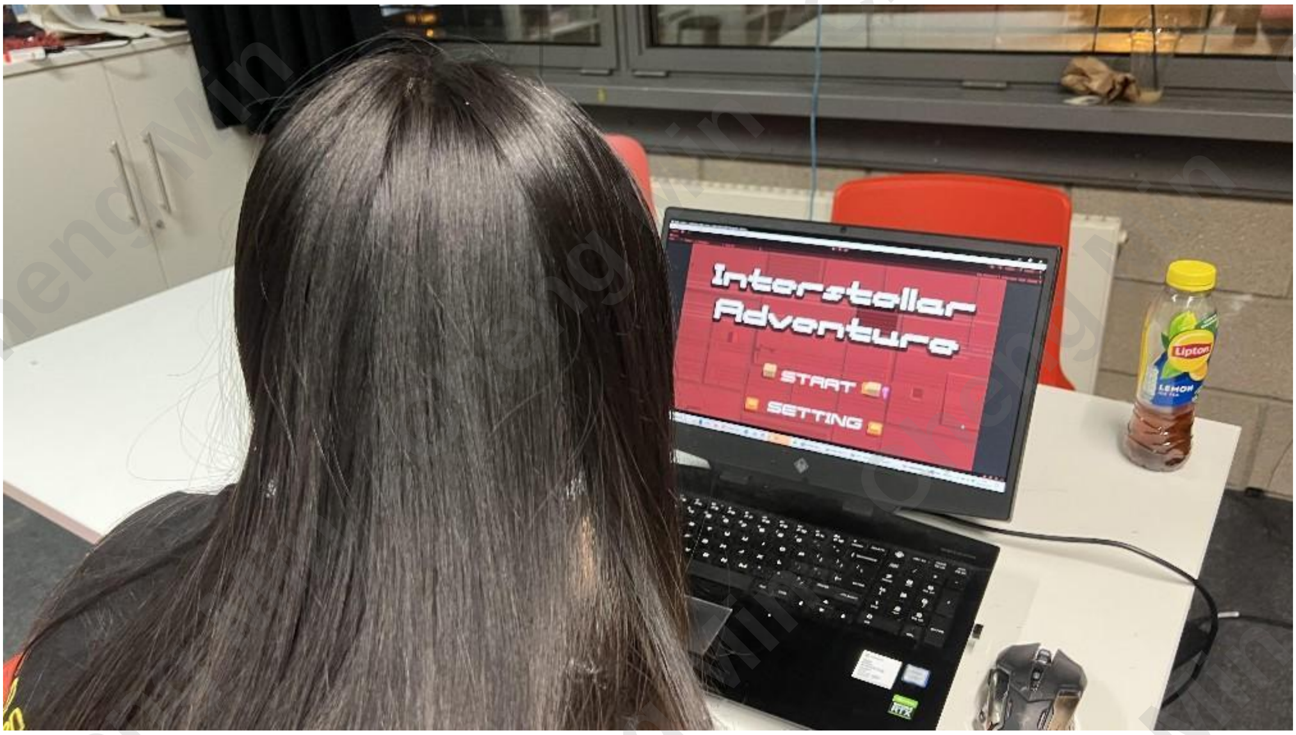


The number of crystals in the map

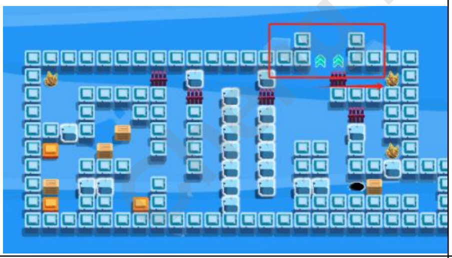
Level			Requirement		How many times encourage players to use	total
1	2		Ice breaker	2	3	6
2	4		Spike Walk	3	3	9
3	2		Pull Box	5	3	15
4	4		Teleport	7	3	21
5	4					
6	4					
7	3					
8	3					
9	3					
10	4					
11	4					
12	4					
13	4					
14	2					
15	3					
16	4					
17	4					
18	1					
19	4					
20	3					
21	5					
22	4					
23	0					
total	75			Anticipated Player Demands		51

Playtesting (11.21)

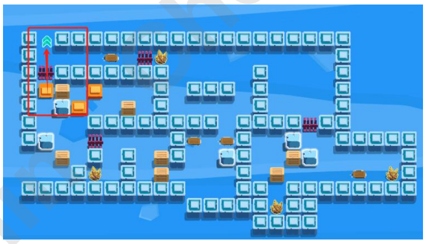
Player 1	**Yiqing**

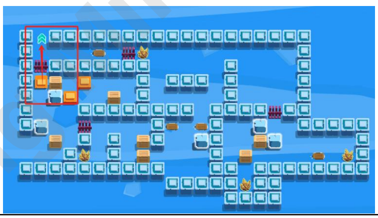


Level	**Player Behaviour**	**Analysis**
Tutorial	The first three levels do not involve collecting all the crystals, even though they are relatively simple.	Perhaps there's a lack of awareness regarding the significance of crystals. Consider implementing a mandatory requirement to collect all crystals in the tutorial section before advancing to the next level.
	Can players be allowed to discover that they cannot pass through the spikes first, and then inform them that they can use a skill to overcome this obstacle?	Adjust the triggering sequence of the guidance dialogue?
Ice Break Level	Enjoy breaking all the ice blocks.	Breaking ice blocks can provide a satisfying sensation for players.

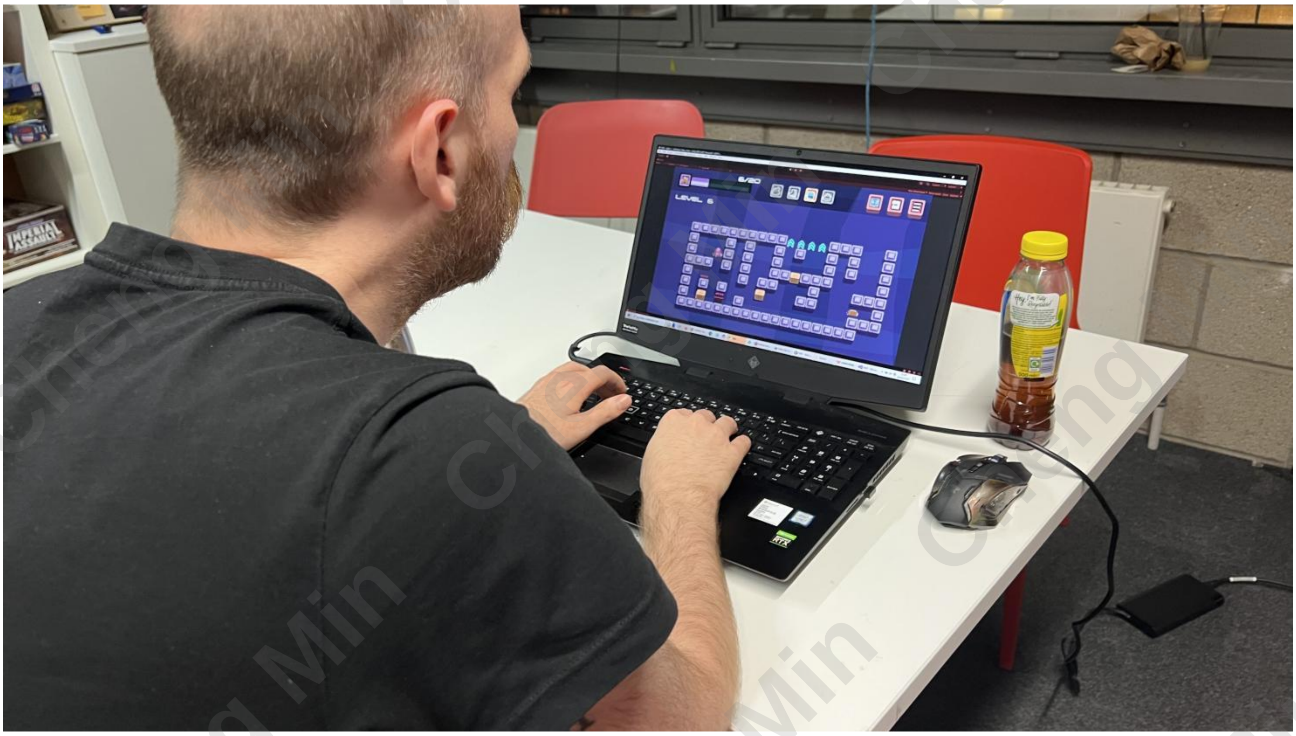
		Consider placing multiple ice blocks consecutively in the ice block obstacle sections to enhance this experience.
Level 9	The paths to the exit and crystal collection are too close, leading to accidental triggers. 	Move the exit upwards.
Level 11	Incomplete crystal collection; encountered insufficient crystals when updating skills for the first time, resulting in an inability to proceed with the game.	Place a sufficient number of crystals in each level where skills need to be updated to ensure players can complete the skill updating requirements.
UI		For the skill page, it would be beneficial to add a crystal icon next to the required quantity to make it clearer; currently, it appears more like an upgrade rather than a quantity requirement.
	The skill list and panel seem a bit confusing; it's not entirely clear how to operate them.	text-based dialogue explaining the usage will be incorporated into the tutorial sections.

Player 2	**Jiage**	
Level	**Player Behaviour**	**Analysis**
Tutorial	The "Z" key for undoing	Include a tutorial for the "Z" key

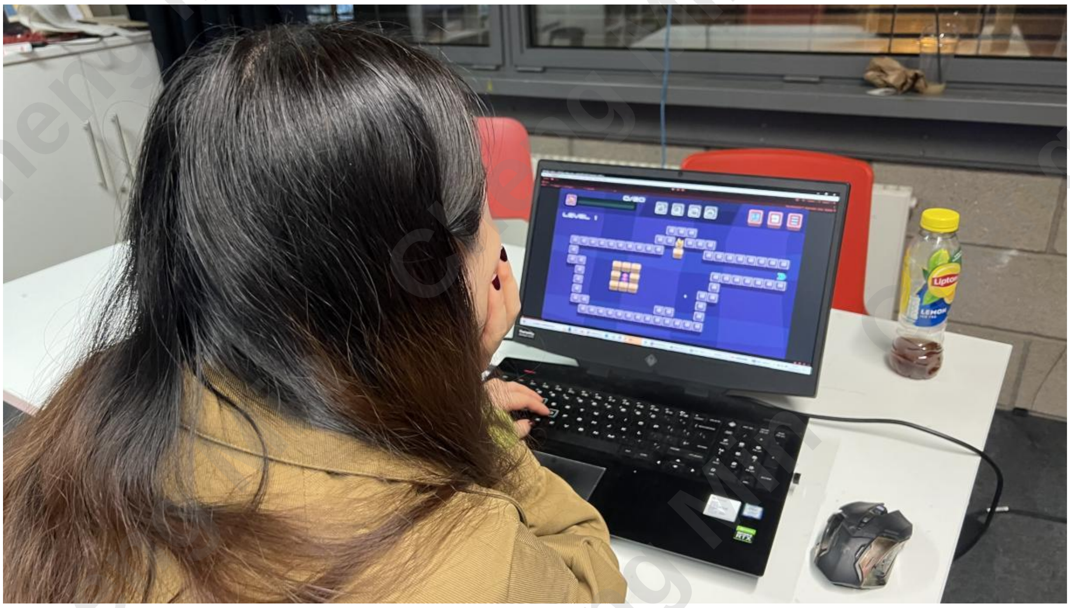
	actions requires a tutorial; otherwise, it's not clear to the player.	functionality.
Level 11	Surprised and unprepared when prompted to submit skills.	Same like Yiqing
	It's possible to directly push a box to the exit to advance to the next level. 	

Player 3	**Zixin**	
		
Level	**Player Behaviour**	**Analysis**
11	It's possible to slip away directly through the fence. 	Remove this gate.

UI	Clicked on "confirm" during skill exchange, unsure if it had any effect.	After confirming, highlight the skill or dim the "confirm" button to indicate its effectiveness.
	For box pulling (K) and teleportation (L), consider displaying the corresponding keys (K and L) on the main interface for user convenience.	

Player 4	David	
		
	Opinion	**Analysis**
Mechanics	Whether to consider mandating that players must collect all crystals to proceed to the next level is worth contemplating.	Since the game currently lacks difficulty levels, enforcing the collection of all crystals might be overly challenging. Perhaps it could be transformed into an achievement instead.

Player 5	Xiaomin	
		
	Opinion	**Analysis**
Mechanics	It's delightful to be able to bypass challenging puzzles in some levels and proceed directly to the next one.	Without a current difficulty grading in the game, this setup allows players who find the game too challenging to still experience a sense of achievement by passing the whole game.
Animation	The character's movement speed feels slightly fast	Reduce the frame rate of movement.

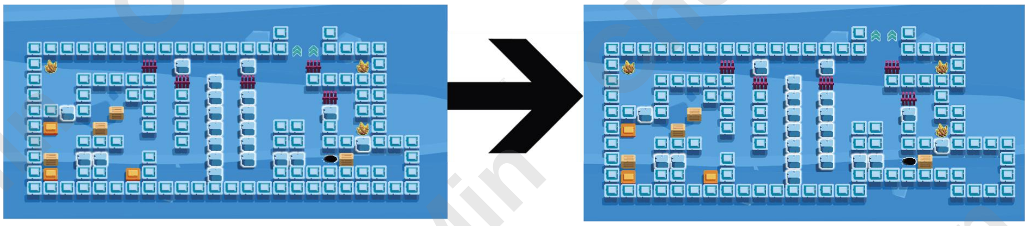
Player 6	Xuan
	

Level	Player Behaviour	Analysis
Tutorial	When filling skills with crystals, it's unclear whether they need to be completely filled to be usable.	Tutorial need to be add
UI	Is it possible to enable the automatic use of crystals when the required quantity is reached, eliminating the need for continuous clicks? Alternatively, consider implementing a one-click fill option.	
Animation	The character's movement speed feels slightly fast	Reduce the frame rate of movement.

Version 2.2

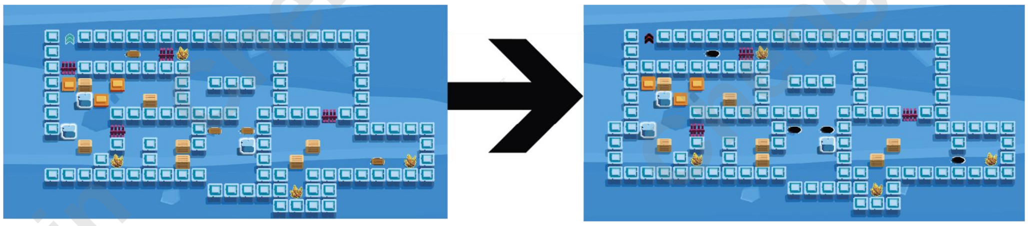
Level Iteration

9



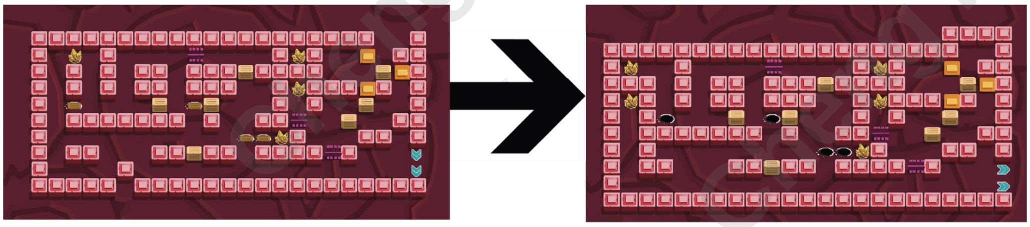
- Move the exit icon upward to prevent accidental triggering by players.

11



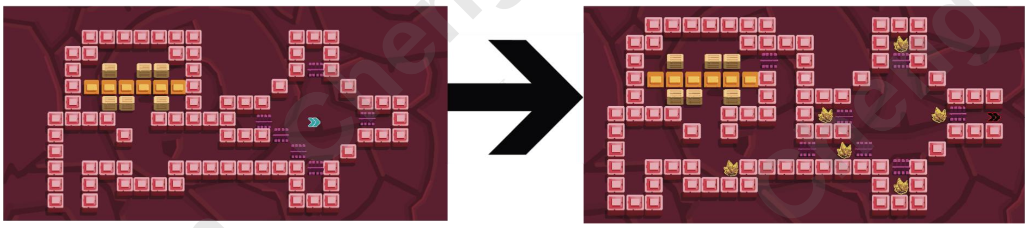
- Remove the gate that allows direct access to the exit to prevent players from getting stuck by pushing the box there, making it impossible to proceed.

22



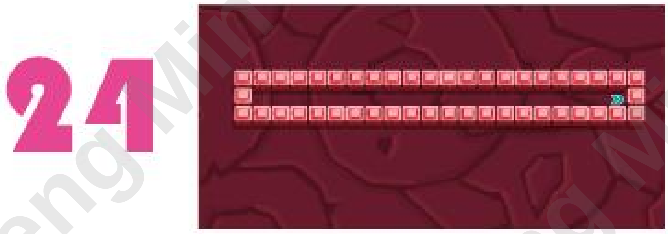
- Modify the arrangement of walls on the left side of the map. Instead of having two paths leading to the same crystal, create two separate paths leading to different crystals, with no interconnection between them.

23



- Add a wall to the right of the switch area to restrict players from pushing the box out of that region, increasing the difficulty.

- Add a pathway above the row of switches to allow players to exit this area. Otherwise, following the intended gameplay, players might get stuck above the switches without a means of leaving.



- Added level 24, connecting it with the ending animation.

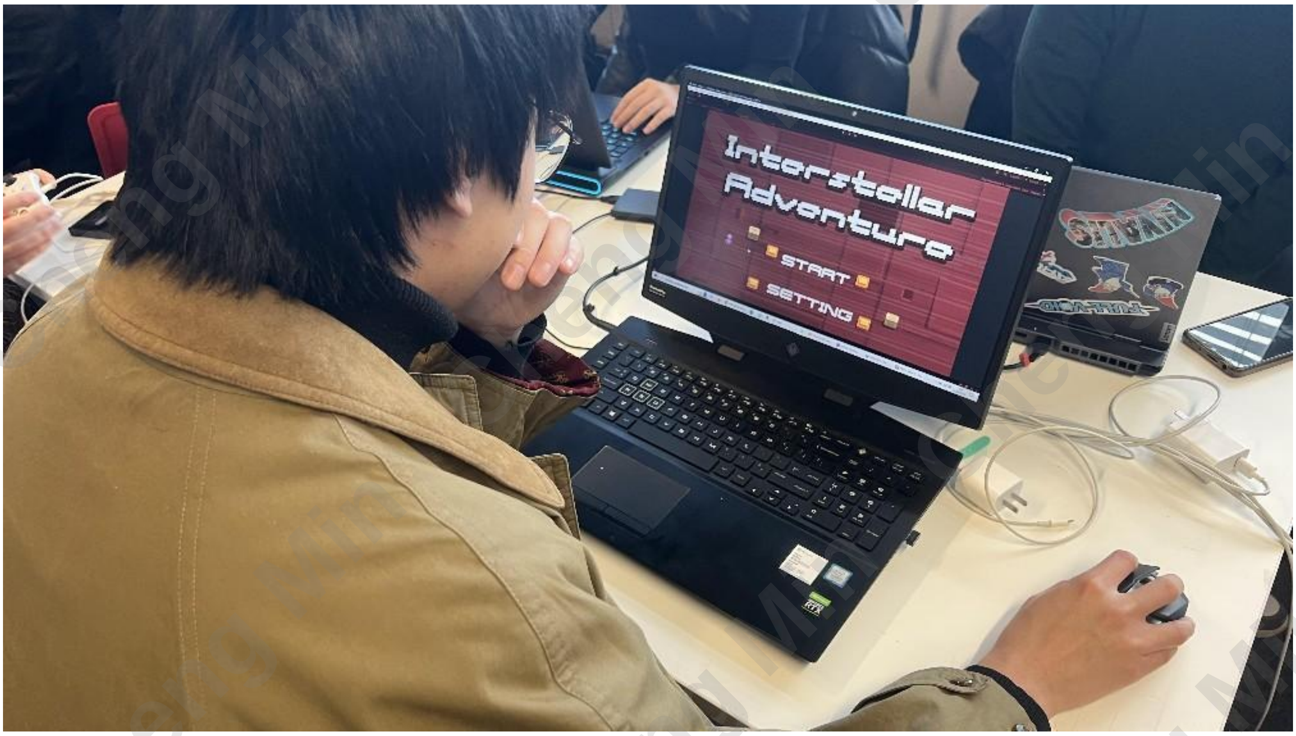
Other Level

- Add surrounding walls at the entrance to prevent players from moving backward from the spawn point and circling around the map edge to reach the endpoint.
- Add a layer of transparent walls around the existing walls to prevent players from using teleportation skills to exit the map.
- Adjust the quantity of crystals required for activation.

Level			requirement		How many times encourage players to use	total
1	1		Ice breaker	4	3	12
2	4		Spike Walk	5	3	15
3	2		Pull Box	7	3	21
4	7	use	Teleport	6	3	18
5	4	use		22		
6	4	use				
7	3	use				
8	2					
9	3					
10	4					
11	4	use				
12	4					
13	4	use				
14	2					
15	6	use				
16	4					
17	4					
18	1	use				
19	7	use				
20	3					
21	5					
22	4	use				
23	6	use				
total	88				Anticipated Player Demands	66

Playtesting (11.24)

Player 1	**Shef**
Level	**Opinion**
16-19	Buttons and orange walls look quite similar.
Art	The box doesn't completely cover the button when pushed onto it; there's a slight misalignment.
UI	Can the functionality of refreshing the entire page be added with the "R" key?

Player 2	**Wenda**
	
Opinion	
● The teleportation feature feels really cool. ● It's a bit challenging; having a save function would be helpful to study at a more relaxed pace.	

Critical Reflection

1. RPG and Narrative

I aimed to create an enjoyable game, and while Sokoban is known for its simple yet challenging gameplay, the market is saturated with similar titles. Seeking innovation, I infused RPG elements into Sokoban to bring a fresh and engaging twist to the classic box-scrambling concept.

This integration offers several advantages, such as:

- In-depth Experience: RPG elements can enhance the game with a storyline, character development, and a rich world background, immersing players more deeply in the gaming experience.
- Motivational Incentives: Role-playing elements can provide tasks, goals, and rewards, motivating players to complete levels and challenges.
- Personalized Gameplay: Players may have the ability to customize characters, choose skills, or make decisions, personalizing the gaming experience.
- Long-Term Appeal: RPG elements often include character upgrades and long-term objectives, increasing the game's longevity and appeal.

However, challenges include increased complexity:

- Increased Complexity: Introducing RPG elements may make the game more complex, potentially making it challenging for Sokoban players who prefer simple puzzle-solving.
- Game Balancing Challenges: Integrating different elements may pose challenges in balancing the game, requiring extra effort to ensure coordination between RPG and Sokoban elements.
- Learning Curve: Familiarizing new players with RPG elements may necessitate additional learning, leading to a potentially steep learning curve for the game.
- Loss of Originality: If RPG elements overshadow Sokoban's classic puzzle-solving aspects, there's a risk of diluting the original gaming experience.

In navigating these considerations, I aim to strike a balance that preserves the essence of Sokoban's simplicity while introducing exciting RPG dynamics. This approach ensures a game that not only captures the timeless challenge of Sokoban but also offers a novel and immersive experience for players.

2. Level Update & Feedback loops

The implemented system is designed to uphold a delicate equilibrium, ensuring that the presented challenges strike the right balance between difficulty and achievability. This approach is aimed at maintaining player interest and immersion over time. The fine-tuning of difficulty levels contributes to a personalized gaming experience, accommodating players of diverse skill levels and ultimately elevating overall satisfaction.

However, it's important to recognize that the incorporation of negative feedback loops necessitates careful consideration and meticulous planning. In the current state of the game, the mechanics are relatively complex, and the tutorial lacks clarity, resulting in higher learning costs for players and potential confusion.

The addition of multiple feedback loops serves to enrich the gameplay, offering players a variety of challenges and strategic choices. While this complexity adds depth, it may prove overwhelming for players unfamiliar with puzzle games or RPGs, potentially deterring some from fully enjoying the experience due to an increased learning curve. Therefore, striking a balance in complexity is crucial for optimizing player engagement

3. Future Enhancement Proposal

In our future development plans, we have outlined three key aspects that we can consider:

Achievement

The box-pushing game is highly suitable for incorporating an achievement system. The achievement system can offer players additional challenges and objectives, enhancing the game's replay value. By completing various achievements, players can showcase their skills and accomplishments in the game, fostering their interest. Achievements can encompass various aspects, such as

- completing all levels without using any skills
- accomplishing specific challenges using particular skills
- solving puzzles within a specified number of moves
- Successfully collect all crystals on the map

This system not only adds depth to the game but also provides players with more enjoyment and motivation.

Save Function

Adding a save function to a Sokoban game has several benefits:

- **Improved User Experience:** The save function allows players to save their progress at any time, preventing the loss of game data due to unexpected events. This enhances the overall gaming experience, making players more willing to invest time and effort.
- **Practicality:** The save function enables players to exit the game in the middle and return to a previous point in their progress without starting over. This is especially crucial for players with limited gaming time.
- **Trying Different Strategies:** Players can experiment with different strategies and approaches, knowing that if they fail, they can return to a previous save point to try again without losing all progress, especially for the different sequence to submit the skills.
- **Reduced Frustration:** The save function alleviates the impact of failure in the game since players know they can return to a previous state when needed. This helps reduce frustration, making the gaming experience more enjoyable.
- **Exploring Game Content:** The save function makes it easier for players to explore different areas and levels of the game, as they can return to and explore previous locations when desired.

Narrative & Art

Upon reviewing the game as a whole, it appears that the narrative flow is not yet seamless, and certain information is still not adequately conveyed. Going forward, it may be beneficial to consider refining the narrative presentation, possibly incorporating storytelling elements into the tutorial to make it more engaging.

Additionally, improvements in the artistic aspects could be explored, such as enhancing the level backgrounds. Currently, they only have a basic color scheme, but according to the initial concept, incorporating specific environmental features of the planets could add more depth, tying them in with the props within each level.